A Cognitive Account of Collective Emergence in Design

Vignettes from a study of real-time collaboration in engineering design

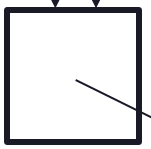
Ben Shaw • IBM Research • Royal College of Art, London

Emergence

- Novelty that is non-obvious and non-predictable
- A recognizable whole or coherent totality that is distinct and qualitatively different from the constituents that preceded it

Design emergence
visual cognition
[Oxman]

Collaborative emergence
jazz and theatrical improvisation
[Sawyer]



Collective storymaking
ethnographic studies of engineering design
[Bucciarelli]

Connections in design conversation
analysis and coding
[Goldschmidt, Clark]

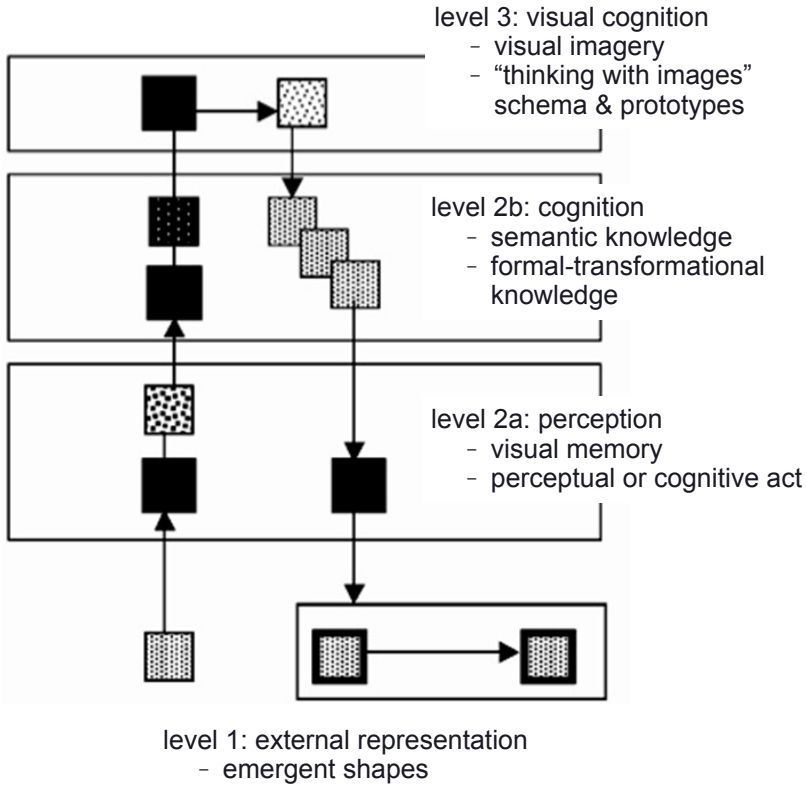
Design emergence
[Oxman (2002)]

shape grammars

perceptual emergence

semantic and transformational knowledge

visual re-cognition



[Oxman (2002) "The Thinking Eye: Visual Re-Cognition in Design Emergence"]

Collaborative emergence

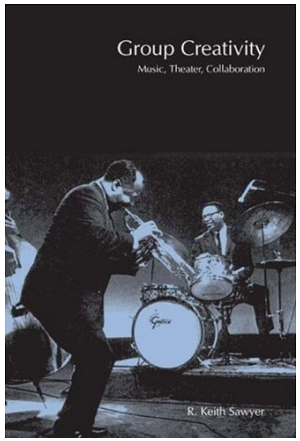
[Sawyer 2002, 2003a,b]

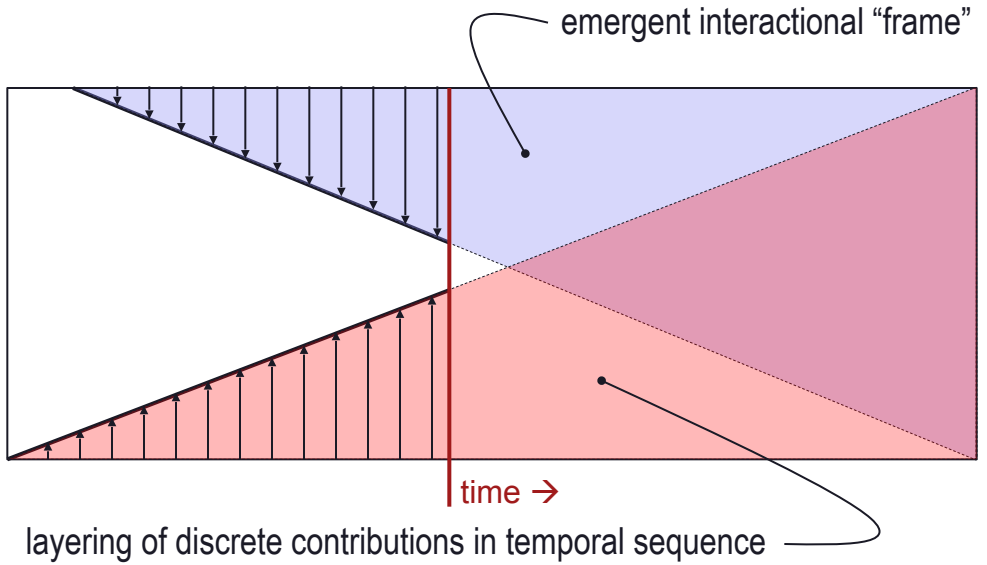
constructive engagement and mutual appropriation

dynamic balance between coherence & inventiveness

upward causation vs. downward constraint

“readymades”





Collective storymaking
[Bucciarelli 1994]

object worlds

- physical artifacts, phenomena, symbolic & mathematical relationships

engineering rhetoric or “cosmology”

- abstraction, sparseness & reduction
- qualified estimation & measurement
- constraint & conservation principles (i.e. tradeoffs)
- cause-and-effect chaining
- hierarchical decomposition

Collective storymaking
[Bucciarelli 1994]

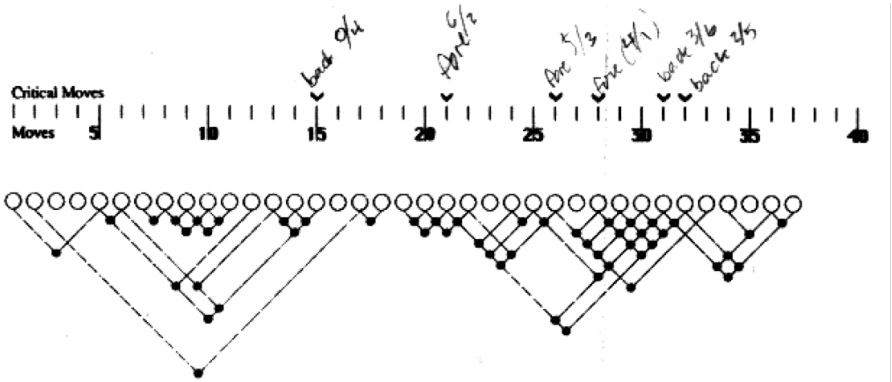
"More than a common vocabulary and syntax, it is a web of tacit understandings of what is to be considered an honorable claim, a significant conjecture, a valid 'proof,' or a laughing matter. It is an accepted rhetoric for describing, proposing, critiquing, and disposing that girds all design conversation, fixing what constitutes a true and useful account."

(Bucciarelli 1994, p. 83)

Connections in design conversation

structural vs. categorical coding

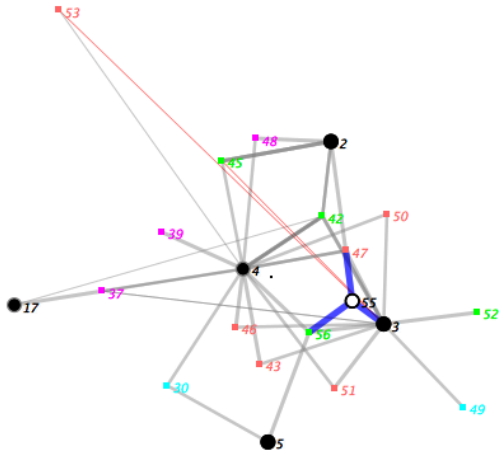
- connections between “ideas’
- Goldschmidt: linkography



Linkograph example. (Goldschmidt, 1992)

Grice: cooperative principle

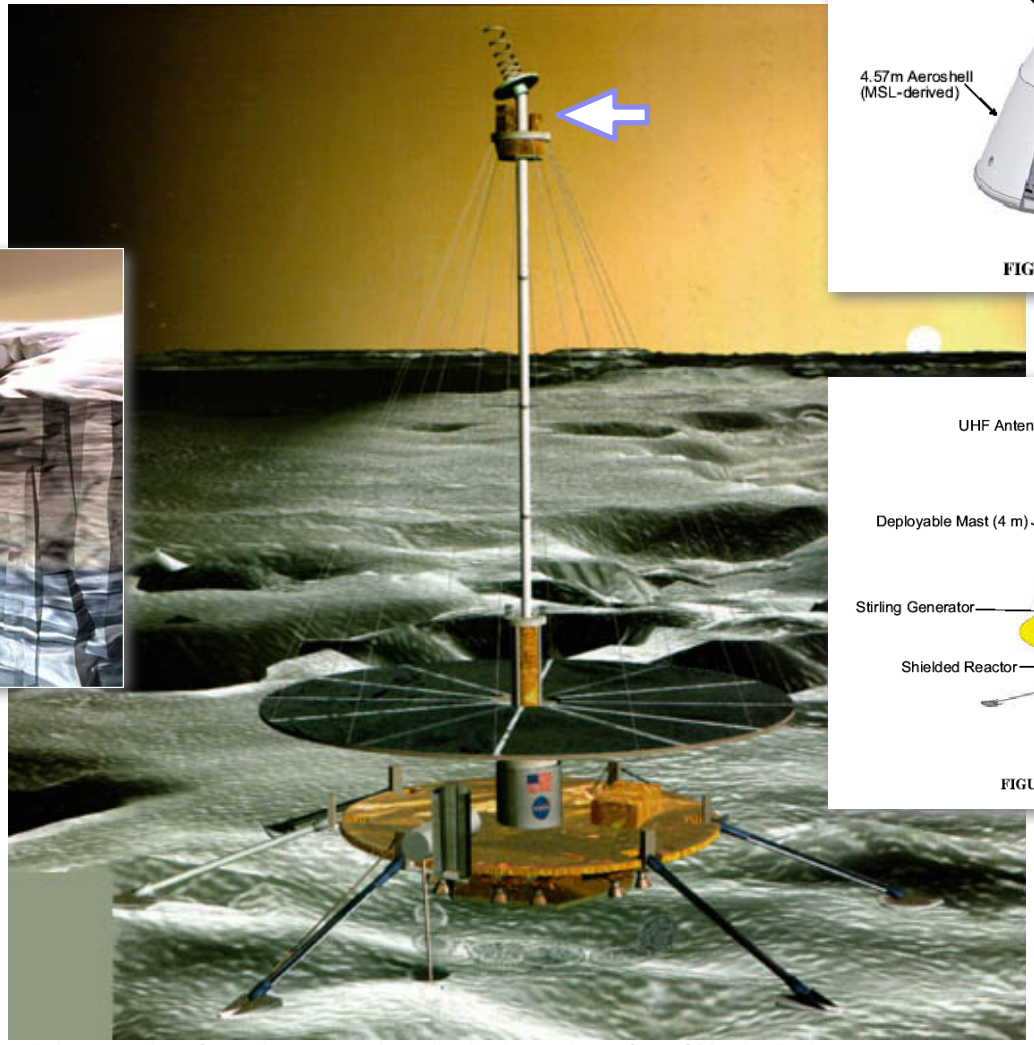
Clark: layers, levels & tracks



Actor-discourse network. (Shaw, 2007)

Designing a cryobot lander





artist's conception of a nuclear-powered cryobot lander on the surface of Mars' polar ice cap

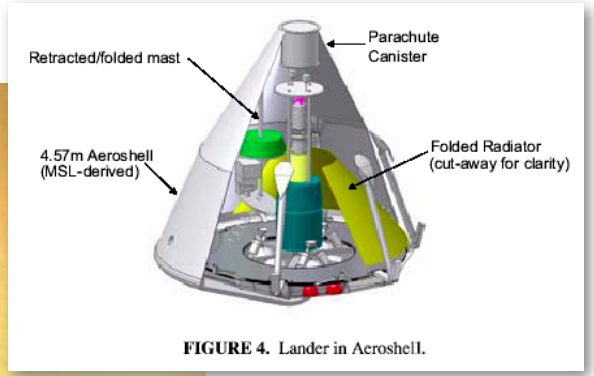


FIGURE 4. Lander in Aeroshell.

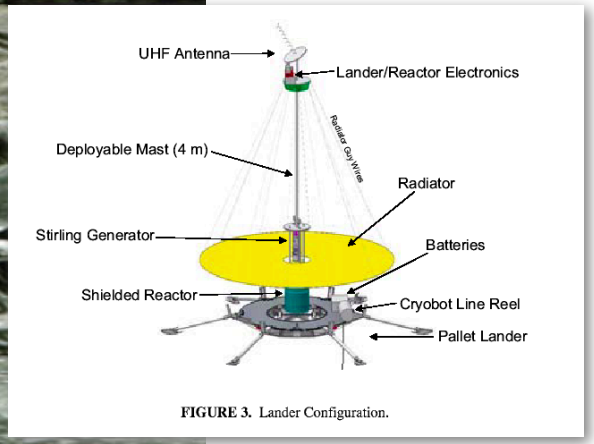


FIGURE 3. Lander Configuration.

Reconstruct Mars' climate history going back millions of years!

Find evidence of life?

But you need kilowatts of power ...

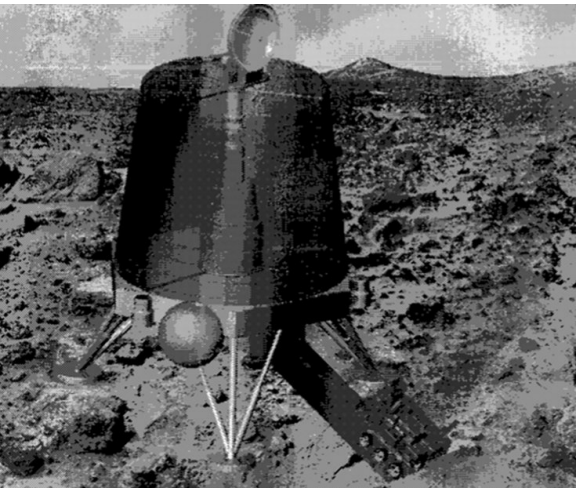
Radiator size?

Sensitive electronics & shielding mass?

Launch windows and landing site selection?

Telecommunications?

Electronics on a mast



Where will we put the sensitive electronics?

A

Melt them into the ice for protection

Put all the sensitive electronics on an extensible boom

B

Telescope *up* instead of out

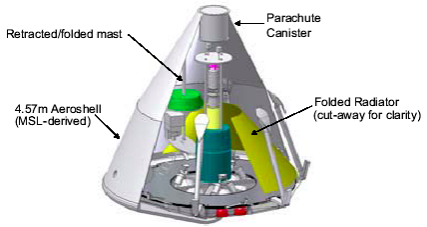
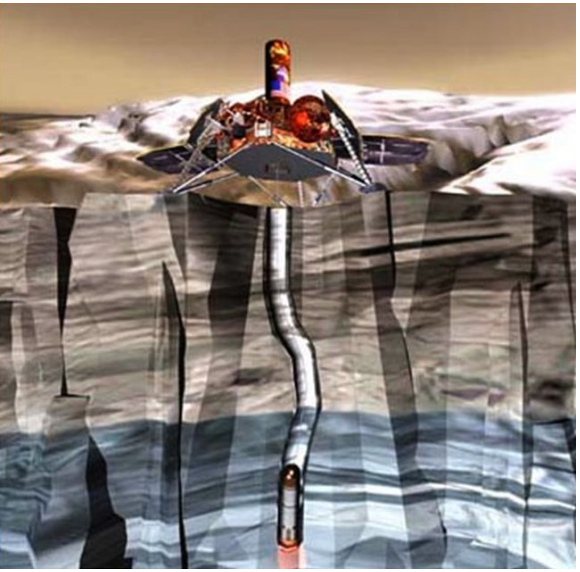


FIGURE 4. Lander in Aeroshell.



How will we fit the radiator in the aeroshell?

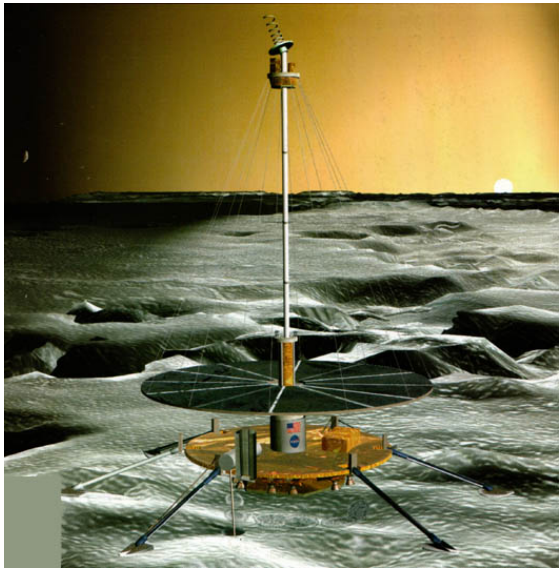
{alternate config 1}

{alternate config 2}

Horizontal disk, facing up

C

Fold down, raise with mast



Protecting sensitive electronics



contribution	*domain/conceptual knowledge*	*image-schematic content*
HL: Cryobot itself shielded in ice, but where do we put other sensitive electronics?	radiation damages electronics water attenuates radiation	[ice protects] [location: on lander] [different boxes]
MW: Mini-rover to move electronics 100yds away.	distance attenuates radiation	[distance: 100yds]
HL, MW: Problems w/mini-rover on snow and ice.	nature of Martian terrain	[traversing terrain]
LC: What about placing cryobot on telescoping arm?	spacecraft design precedents	[telescoping arm] [distance: few meters]
HL: Yeah, we would like to place cryobot in different spots.	science objectives, stratigraphy	[arm moving]
ZD: Is the separation great enough to be interesting?	stratigraphy	
HL: A hundred meters *is* scientifically interesting.	stratigraphy, citing an expert	[distance: 100m]
MW: Can you have two cryobots? You could shoot them out from a small canon	highly shock resistant spacecraft design precedents	[distance: 50m] [object flying]
ZD: It would have to land tip down	cryobot design	[object landing]
MW: After heating cryobot would tend to right itself	cryobot design, object reorientation during sinking	[melting into ice]
A **LC**: What about melting package of sensitive electronics into ice to shield?	packaging spacecraft electronics	[melting into ice] [ice protects] [all electronics]

T

T

T

C

C = combination; T = transformation

Protecting sensitive electronics





	contribution	*domain/conceptual knowledge*	*image-schematic content*
	HL: What about putting all the electronics on an extensible boom?	spacecraft design precedents distance attenuates radiation	**[all electronics]** **[telescoping boom]**
	HL, MW: A boom with skis? Multiple booms?	engineering design solutions	**[cantelievered weight; balance, support]** **[pushing snow]**
	MW: A boom that unrolls like a party whistle?	nature of Martian terrain	**[horizontal movement]**
	ZD: What if you land at an angle, or right next to a rock?	nature of Martian terrain	**[encountering obstruction]**
B	**HL**: What if you telescope *up* instead of out!	engineering design solutions	**[telescope up]**

C

T

C = combination; T = transformation

Radiator configuration



	contribution	*domain/conceptual knowledge*	*image-schematic content*	
	CAD representation with unarticulated horizontal disk at specified height above reactor		[fold like {a familiar object}]	T
C	**HL, IE** (separately): Instead of folding the way we discussed, we could fold the radiator *down* and raise it with the mast!	precedents for spacecraft articulation & deployment physical constraint of aeroshell volume	[fold down] [raise with mast] [open like {a different familiar object}]	

C = combination; T = transformation

Summary

novel outcomes result of essentially collective process

conversation proceeds primarily through schema extraction and simple transformation (c.f. Oxman's account)

overall emergent outcomes were a result of layering contributions of multiple participants (in the manner described by Sawyer)

Collective emergence

Social Interaction
- communicative act

- collective storymaking
- narrative transformation

Visual Cognition

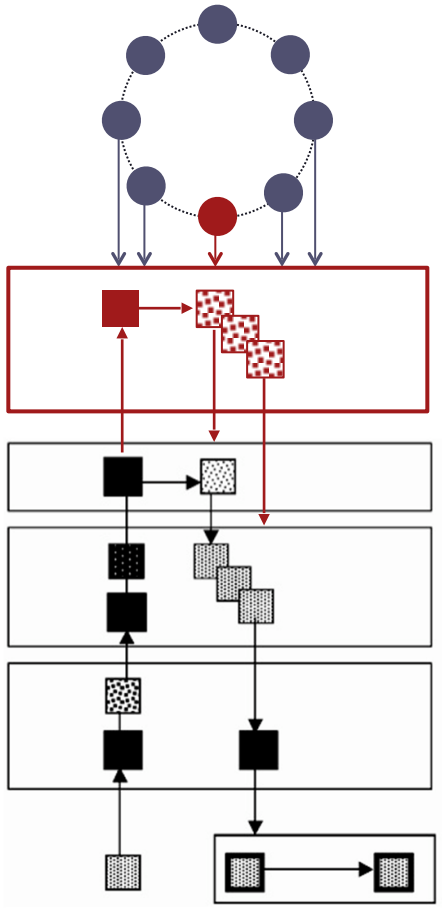
- visual imagery
- "thinking with images"
- schema and prototypes

Cognition
- semantic knowledge

- formal-transformational knowledge

Perception
- visual memory

- emergent perceptual/ cognitive product



further work

creative conversations in other domains?

- schemas common or different, across e.g. business, accounting, manufacturing, server & database administration?
- requisite knowledge & common ground necessary to engage in creative conversation?

schema transformations queued by visual representations vs. those in talk alone?

impact of social dynamics and hierarchy?

Adjacency

C-K design theory: an advanced formulation

Original Paper | Published: 19 August 2008
Volume 19, pages 181–192 (2009) Cite this article



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Abstract

C-K theory is a unified Design theory and was first introduced in 2003 (Hatchuel and Weil 2003). The name "C-K theory" reflects the assumption that Design can be modelled as the interplay between *two interdependent spaces* with different structures and logics: the space of concepts (C) and the space of knowledge (K). Both pragmatic views of Design and existing Design theories define Design as a dynamic mapping process between required functions and selected structures. However, dynamic mapping is not sufficient to describe the *generation of new objects and new knowledge* which are distinctive features of Design. We show that C-K theory captures such generation and offers a rigorous definition of Design. This is illustrated with an example: the design of Magnesium-CO_2 engines for Mars explorations. Using C-K theory we also discuss Braha and Reich's topological structures for design modelling (Braha and Reich 2003). We interpret this approach as special assumptions about the stability of objects in space K. Combining C-K theory and Braha and Reich's models opens new areas for research about *knowledge structures* in Design theories. These findings confirm the analytical and interpretative power of C-K theory.

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